Keep the 5 engineering snippets you need. Close the 37 hoarded tabs you will never read.

Folivora (Latin: *folium + vorare*, "the page devourer") is an open-weight AI browser workspace and Chrome extension that inspects your open tabs in Browserbase Cloud Chromium, predicts calibrated P10/P50/P90 utility quantiles with Prior Labs TabPFN, extracts section-anchored takeaways behind deterministic verification gates with Google Gemma, and speaks a 20-second executive briefing via ElevenLabs.

FALSE-CLOSE SAFETY

0.36%

TabPFN P90 guardrail across 3,600 developer tabs (720 unseen test rows).

FAST-LANE TRIAGE

~1.3s

LLM-free deterministic cards on a laptop CPU; preset repeats served from cache in <30ms.

FACT-GATE PASS RATE

98.3%

59/60 verified spoken digests behind programmatic digit & anchor validation.

Built solo by a systems engineer who ships bare-metal, open-source tools to production.

Omar Afifi is a full-stack systems and AI engineer who built Folivora out of firsthand frustration after watching 40+ documentation and debugging tabs drain system RAM and focus while building production open-source software:

📦 Shrinkeo (Microsoft Store)

Open-source, GPU-accelerated Windows media compression studio published on the official **Microsoft Store**.

- → Hardware-accelerated NVENC, AMD AMF & Intel QSV encoding
- → Exact MB/KB target sizing via 2-Pass VBR optimization
- → 100% offline privacy, EXIF scrubber, and 44 localized languages

`Microsoft Store Live` `GPL v3 Open Source`

⚡ Daphq (Raw TCP Engine)

Open-source, bare-metal cross-platform file transfer engine for Windows & Android that bypasses HTTP overhead entirely.

- → Raw TCP sockets with `SocketOption.tcpNoDelay` & UDP watchdog discovery
- → Zero-copy `Uint8List.sublistView` boundary slicing (0 OOM crashes)
- → Saturates **~93% of physical Wi-Fi bandwidth at 90+ MB/s**

`90+ MB/s Verified` `MIT Open Source`

Browser tabs have become an anxious external memory bank for 1B+ knowledge workers.

Engineers and researchers don't keep 42 browser tabs open for three weeks to read 5,000-word essays. We keep them open because **5 of those tabs contain a 3-line CLI flag, SQL snippet, or benchmark number** we are afraid to lose.

❌ Why Current Solutions Fail

- **Bookmark & Tab Managers (OneTab, Pocket):** Turn clutter in the tab bar into a static graveyard of links you still never read.
- **Single-Tab AI Chatbars (Copilot, ChatGPT Sidebars):** Summarize one tab at a time, strip out CLI flags/code blocks, and cannot triage 40 tabs simultaneously.
- **Unverified LLM Wrappers:** Hallucinate version numbers, invent flags, and provide zero calibrated confidence on whether a tab is safe to close.
- **Cognitive & RAM Tax:** 40+ active Chromium renderers consume 4–8 GB of RAM and fragment deep engineering focus.

✅ What Knowledge Workers Actually Need

- **Calibrated Triage with Uncertainty Bounds:** A statistical guarantee (`P10 / P50 / P90`) that closing a tab won't throw away critical reference material.
- **Section-Anchored Extraction:** Exact `Section Heading — Fact` takeaways plus preserved copyable code blocks (`pre/code`).
- **Zero-Prompt & Topic-Focused Modes:** Summarize every open tab immediately, or filter by voice/text intent (e.g., *"Postgres HNSW"*) and hibernate unrelated tabs in one click.
- **Eyes-Free Executive Briefing:** A 20-second spoken audio summary verified digit-by-digit against ground truth.

Meet Folivora: Deterministic LLM-free triage + grounded open-weight AI polish.

Folivora decouples instant structural triage from background LLM synthesis—rendering interactive dossiers in ~1.3s warm (~30ms from the preset cache) while upgrading prose asynchronously behind strict mathematical verification gates.

STAGE 01 · CAPTURE

🌐 Browserbase CDP + Extension

Captures open window URLs via Manifest V3 Chrome Extension or opens live pages in Browserbase Cloud Chromium over raw CDP WebSockets.

STAGE 02 · SIEVE

📊 Prior Labs TabPFN v2

Scores 7 structural & intent features on CPU to predict calibrated `P10` / `P50` / `P90` quantiles. Enforces `P90 < 45` safety gate.

STAGE 03 · FAST LANE

⚡ Instant Dossier (Zero LLM)

Immediately renders sanitized section-anchored takeaways, copyable code blocks, reading time saved, and keep/close verdicts.

STAGE 04 · GROUNDED AI

💎 Gemma 4 + Gemini Flash Lite

Upgrades executive brief (`check_output()` digit gate) & polishes top 8 tab cards against bounded `E#/C#` evidence packs.

STAGE 05 · VOICE & TRACE

🎙️ ElevenLabs + Sentry

Scribe (`scribe_v1`) voice intent input (with immediate remote delete), Flash v2.5 spoken briefing, and scrubbed Sentry OpenTelemetry spans.

Three ways to experience Folivora live right now.

Every feature is live and testable in the browser at folivora.tech with zero login walls.

DEMO PATH 01

🏆 65-Tab DEV Hall of Fame Pile

Triage all 65 featured DEV Community articles (600+ minutes of reading) in one deterministic LLM-free pass (~1.6s warm; repeat loads ~30ms from cache). Filter by any topic (e.g., `Gemini`, `Rust`, `Docker`, `Flutter`) and watch TabPFN isolate the exact matching articles and jump straight to the matching H2/H3 section.

`65 Real Articles` `LLM-Free Triage`

DEMO PATH 02

🔗 Paste Any Live URLs + Cloud CDP

Paste up to 65 live URLs from your own browser—or toggle **Live Browserbase Cloud Chromium** to inspect live DOMs over raw CDP WebSockets with an embedded live session replay link and automatic session release.

`Browserbase CDP` `SSRF Protected`

DEMO PATH 03

🧩 One-Click Chrome Extension

Download `/download-extension.zip` directly from the navbar (pre-configured to `https://folivora.tech`). Scan your current browser window, hibernate unrelated tabs to **0 MB RAM** via `chrome.tabs.discard()`, or close them with a 1-click Undo snapshot.

`Manifest V3` `1-Click Undo`

Engineered with measurable ML rigor—not prompt-and-pray.

Why use Prior Labs' TabPFN v2 quantile distribution instead of a standard point-estimate classifier? Because in tab triage, **falsely closing one critical reference tab is 10x worse than keeping an extra tab open**.

MODEL / DECISION POLICY (720 UNSEEN TEST TABS)	RMSE ↓	SPEARMAN P ↑	FALSE-CLOSE RATE ↓	RECALL ON MUST-KEEP ↑
🏆 **Prior Labs TabPFN v2 (P50 + P90 < 45 Guardrail — Shipped)**	**6.83**	**0.9557**	**0.36% (1 / 276)**	**99.64%**
TabPFN Point-Only Policy (P50 < 42, ignoring P90 uncertainty)	6.83	0.9557	1.81% (5x riskier)	98.19%
HistGradientBoostingRegressor (3-Model Quantile Ensemble)	7.26	0.9498	1.09%	98.91%
RandomForestRegressor (100 Trees)	7.45	0.9469	2.17%	97.83%
Heuristic Word-Count + Age Rule	28.86	0.4517	19.20%	80.80%

🛡️ 59/60 (98.3%) Digest Gate

`check_output()` parses every number in Gemma's prose and rejects any rounded, invented, or swapped digit with a 1-retry correction loop.

🔪 11/11 Golden Summary Gates

`tab-summary-validation-v3` enforces exact section anchors, grounded numbers/units, zero first-person voice, and anti-verbatim synthesis.

🔒 0.0% Red-Team ASR (24/24)

Built for enterprise-grade privacy and zero-trust web ingestion.

Every external web page inspected by an AI browser agent is untrusted input. Folivora treats web DOMs as adversarial by default and isolates user sessions from end to end.

🔐 Multi-Layer Privacy & Isolation

- → **Private Ephemeral Job Store:** Custom user URLs are isolated in a per-job TTL/LRU store (`15m` expiry) and never written to shared caches or serialized back in raw API responses.
- → **Transcribe-and-Forget Voice Privacy:** Voice intents sent to ElevenLabs Scribe (`scribe_v1`) trigger an immediate remote `DELETE /v1/speech-to-text/transcripts/{id}` in a `finally` block.
- → **SSRF & Network Hardening:** Blocks RFC1918 private IPs, loopback, link-local metadata endpoints, and credentialed URLs on every redirect hop.

🔭 Observability & Cost Efficiency

- → **Sentry OpenTelemetry GenAI Tracing:** Tracks latency, token throughput, and estimated cost across `browserbase.inspect_tabs`, `tabpfn.predict_quantiles`, and `gemma.digest` spans.
- → **Regex Secret Scrubber:** Redacts `api_key`, `tabpfn_token`, and `signingKey` while preserving `gen_ai.usage.input_tokens` and `output_tokens`.
- → **Sub-Cent Unit Economics:** Pre-baked CPU TabPFN weights ($0 API cost per triage) + bounded evidence packs keep full 8-tab AI polish under $0.002 per batch.

If this works, how big could it get?

The web browser is the primary operating system for over **1 billion desktop knowledge workers**—yet tab management hasn't fundamentally evolved since tabbed browsing was invented in 1998.

BEACHHEAD (SOM)

30M+

Software & AI Engineers

Developers and technical researchers who juggle 40+ GitHub issues, StackOverflow threads, cloud docs, and arXiv papers daily and value deterministic accuracy and preserved code snippets.

EXPANSION (SAM)

250M+

Technical Knowledge Workers

Product managers, security analysts, financial researchers, and graduate students conducting multi-source synthesis across dozens of browser tabs every day.

GLOBAL VISION (TAM)

1B+

Desktop Browser Users

Becoming the intelligent memory, triage, and voice-briefing layer across Chrome, Brave, Edge, Arc, and enterprise browser environments worldwide.

Day-Zero live build with a clear Open-Core SaaS monetization engine.

Folivora is 100% bootstrapped ($0 external funding, pre-revenue, launched October 2026) with a complete working web workspace and Chrome extension live on **folivora.tech**.

TIER 01 · COMMUNITY

🌱 Open-Source / Free

$0 / mo

- → Unlimited local deterministic tab triage (LLM-free)
- → Chrome Extension tab hibernation & 1-click undo
- → Self-hosted Docker & local Ollama support

TIER 02 · INDIVIDUAL PRO

⚡ Folivora Pro

$12 / mo

- → Cloud Browserbase CDP rendering for JS-heavy SPAs
- → Unlimited Gemma 4 & Gemini grounded card polish
- → ElevenLabs voice briefings & searchable tab archive

TIER 03 · ENGINEERING TEAMS

🏢 Folivora Teams

$29 / seat

Turning folivora.tech into the definitive AI browser workspace.

How we are scaling Folivora across browser ecosystems and developer workflows over the next 6 months:

MILESTONE 01 · Q1 2027

🧩 Chrome Web Store & Recall

- → Public launch on Chrome Web Store, Firefox Add-ons & Edge Store
- → Persistent local-first vector recall over every closed tab ("Never fear closing a tab again")
- → Automated daily end-of-day spoken tab cleanup digest

MILESTONE 02 · Q2 2027

💻 IDE & Workflow Bridge

- → VS Code, Cursor & JetBrains MCP integration—inject kept tab snippets directly into coding agents
- → Collaborative team research dossiers on `folivora.tech`
- → Fine-tuned personalization on individual user keep/close feedback

WHY FOLIVORA WINS

🚀 High-Velocity Execution

- → Working, benchmarked product live today on `folivora.tech`
- → Proven solo founder with shipped Microsoft Store & open-source systems
- → Next phase scales Browserbase cloud sessions, Chrome Store distribution, and Pro tier launch